



Universidade Federal do ABC

Simulation Exercise #08: Simulation of Event-Driven Systems

INF301 – System Modeling and Simulation – W08

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Report Guidelines

Use a notebook on the Google Collaboratory platform to generate a report containing any explanations and comments you deem relevant, along with your codes and figures.

The graphs in the figures should be **self-explanatory**, with axis names and data captions. Use a **font size appropriate** for presentation in a document.

The language to be used is Python. However, **the use of pre-built Python libraries is not permitted**, except for those used in the examples.

Submit a **single notebook file in ipynb format**, with the file name in the format SEON_NameSurname.ipynb, where N is the SE number, Name is your first name, and Surname is your last name.

Remember that plagiarism will not be tolerated under any circumstances!

Problem 1

Consider the system presented in Example 4.

- (a) Construct a function that simulates this system, taking as inputs N (number of events), s (number of servers), ρ (system load), and τ (average departure time).
- (b) Using the constructed function, plot a graph showing blocking probability curves as a function of load (ranging from 0.05 to 1 erlang, with steps of 0.05). Consider $N = 200,000$, $\tau = 10$ seconds, with $s \in [1, 2]$. This graph will present two curves, one for each value of s .
- (c) Compare the results obtained in (b) with the theoretical results obtained using the Erlang-B formula, plotting the theoretical and simulated results on the same graph.

Joker Problem (not mandatory)

Consider Example 5, assuming $n = 200,000$; $S = 1$; $L = 2$; $\lambda = 0.8$; $\tau = 1$. Modify the code so that it is possible to obtain:

- (a) The average number of customers in the queue and the average number of customers in the system.
- (b) The average queue time and the average system time.
- (c) From these results, verify the validity of Little's Theorem.



If submitted, the grade for this problem will replace the lowest grade obtained in the SEs, as long as it is higher than the grade to be replaced.